

Naga Satya Phanindra Vallabhaneni

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PROFESSIONAL SUMMARY

May 2026 MS Computer Science graduate with 3+ years of professional software engineering experience: production Python backend services, REST APIs, and data pipelines processing 3.5M+ daily records. Rigorous automated testing (unit, integration, regression) with CI/CD sustaining 99.9% uptime across 14 enterprise environments. Fast learner who ships tested, production-quality code in zero-to-one, high-ownership environments.

TECHNICAL SKILLS

Software Development: Python, REST APIs, Microservices, SQL, JSON, Data Pipeline Design, Workflow Automation

Quality Engineering: Unit, Integration & Regression Testing, Root-Cause Analysis, Debugging, Edge-Case Analysis, Software Triage

DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, Git, CI/CD, AWS, GCP, Linux

Collaboration: Agile/Scrum, Code Reviews, Technical Documentation, Cross-Functional Collaboration, Mentoring

TECHNICAL EXPERIENCE

University of Dayton | Dayton, OH

Dec 2025 – May 2026

AI Solutions Associate

- Built Python-based data ingestion, normalization, and validation pipelines to process and curate 4,400+ complex institutional data records for automated query routing and reliable system behavior.
- Implemented structured Model Context Protocol (MCP) tool schemas and protocol-based routing to support automated validation, connection logging, real-time execution, and systematic issue analysis.
- Evaluated system performance metrics, test coverage, and response behavior, generating analytical performance reports that improved response accuracy to 88% and reduced loss by 50.6%.
- Analyzed edge cases and system behavior across complex data conditions to identify inconsistencies, validate expected outputs, and improve workflow reliability.

Accenture | Hyderabad, India

Oct 2022 – Jul 2024

Infrastructure & Python Engineer

- Developed Python automation scripts and high-throughput data pipelines processing 3.5M+ daily records, automating data workflows and metadata extraction while maintaining high data fidelity.
- Designed and executed comprehensive QA, integration, and regression testing suites across distributed services, reducing production software bugs by 80%.
- Validated inter-service communication, payload schemas, system telemetry, and expected system behavior across 14 enterprise environments, maintaining 99.9% uptime.
- Performed software triage, debugging, fault-tolerance analysis, and root-cause investigation on complex data integration workflows, identifying edge cases and resolving issues affecting system performance.
- Streamlined release and testing workflows using Jenkins, Docker, and GitHub Actions, transitioning deployment workflows from weekly to daily releases.
- Collaborated with engineering teams through code reviews, pair programming, technical troubleshooting, and mentoring junior developers.

LJ Technologies | Hyderabad, India

May 2021 – Jul 2022

Systems & Infrastructure Engineer

- Conducted comprehensive unit, integration, and regression testing across distributed API services to ensure system stability and data consistency.
- Developed scalable Python automation pipelines for data extraction, schema validation, and metadata normalization across backend systems.
- Maintained CI/CD deployment pipelines using Docker, Git, and Jenkins to streamline software release workflows and maintain environment stability.

TECHNICAL PROJECTS

FlyerGPT AI Platform & MCP Agent Infrastructure — Personal Project

- Architected an AI assistant platform and Model Context Protocol (MCP) agent infrastructure using Python, FastMCP, and Azure OpenAI for conversational database access.
- Implemented structured tool schemas, access controls, and connection event logging to validate data queries and ensure reliable system behavior.

Plant Yield and Growth Prediction Using ML and DL Algorithms — IEEE ICEARS 2022

- Developed Python analysis workflows using LSTM, CNN, Random Forest, and SVR models to analyze time-series environmental and growth metrics in controlled environments.
- Evaluated predictive model performance across varying environmental conditions and generated data visualizations and analytical reports to assess model accuracy and sensor data reliability.

Phishing URL Detection Using BERT and Cybersecurity Analysis — Open-source (GitHub)

- Feature-engineered a 12M+ row dataset using Python to build a BERT-based NLP classification pipeline achieving 78% accuracy; open-sourced on GitHub.
- Evaluated model coverage and performance metrics, developing diagnostic scripts to analyze misclassifications and triage edge-case inputs across large datasets.

EDUCATION

Master of Science in Computer Science — University of Dayton, May 2026

Bachelor of Technology in Electrical & Communications Engineering — GRIET, May 2022

CERTIFICATIONS

GitHub Foundations Badge (2025) • NPTEL: Joy of Computing Using Python • Oracle AI Foundations Associate